

June 22, 2026

Admissions Committee
Leiden University
Leiden, Netherlands

Dear Admissions Committee,

I am writing to express my strong interest in the PhD position focusing on Formal Methods in Natural Language Processing. As an AI & Data Science Engineer, my research has consistently focused on bridging the gap between highly parameterized neural networks and rigorous, formal mathematical bounds.

In my recent engineering theses, I focused heavily on deterministic continuous-time models. Specifically, I engineered Medical_MambaTS, leveraging linear-time state-space representations to process complex sequential data with strict mathematical guarantees. Unlike standard transformer attention mechanisms, the state-space formalism allows for exact analytical bounds on sequence generation. Furthermore, my work with Physics-Informed Operator Networks (PI-DeepONet) involved embedding strict formal constraints (differential equations) directly into the neural manifold.

I am highly motivated to apply this background in formal mathematical modeling to the domain of Natural Language Processing. The challenge of integrating formal methods (such as type theory, grammar formalisms, or algebraic structures) with large-scale NLP architectures is exactly the kind of foundational, cross-disciplinary research I wish to pursue at Leiden University.

Thank you for your time and consideration.

Sincerely,
Ismail Aissaoui
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